Vehicle Routing Problem for Urban Freight Transportation: A Review of the Recent Literature



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Abstract This paper presents a literature review of the Vehicle Routing Problem (VRP) for urban freight transportation. After introducing the methodological approach followed for the review, the analysis of the latest bibliography is presented. The fact that the variants of the VRP have grown and similarly have their practices and applications made the VRP popular in the academic literature. Hence, the number of articles published is constantly increasing, and it is difficult to monitor the developments of the research field. In this paper, after considering existing research papers, a review from Scopus scientific database is presented. The taxonomy of the last decade's literature of the VRP for urban freight transportation is given and critical insights are discussed. After the categorization of the articles, the trends of the VRP for urban freight transportation are featured and analyzed. This analysis and the knowledge gained will be used in a forthcoming research project in order to support the application of VRP algorithms in the functionality of a vehicle routing and scheduling information system. This review is a useful tool for academia's and practitioners who study the adoption of VRP algorithms in IT solutions in order to effectively schedule deliveries and routes in city logistics environment.

Keywords Vehicle Routing Problem · Scheduling · Urban Freight Transportation · City logistics · Algorithms · Literature review

1 Introduction

Freight transportation is the delivery of products from distribution centers to several drop points. The scheduling of deliveries and the routing of vehicles in urban areas is affected by traffic conditions which have a significant impact on travel time and

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consequently on delivery efficiency and customer service. Planners have to assign many orders to many vehicles and, for each vehicle, assign a delivery sequence, trying to provide a balance between supplying first class service to customers at an acceptable cost. In doing so, there are a number of additional constraints that must be considered, such as the weight or volume capacity of the vehicle, the total daily time available, loading and unloading times, drivers working hours, different vehicle speeds, traffic congestion, access restrictions, and environmental constraints. Decision making for urban freight transportation involves VRP theory and algorithms in order to model and solve real-life problems in city logistics (Gonzalez-Feliu and Salanova-Grau 2015).

Logistic managers have faced the Vehicle Routing Problem (VRP) in practice, ever since multi-drop freight transportation vehicles were introduced. First attempts to the solution to the VRP have been addressed by various software tools since the 1980s. These attempts have been significantly enhanced by the development of Geographical Information Systems (GIS) and Advanced Planning and Scheduling Systems in the 2000s (Gayialis and Tatsiopoulos 2004). Various software products have been developed to solve the VRP variants, based on the research that has been done. Theoretical research in the field of vehicle routing started with the "Truck Dispatching Problem" posed by Dantzig and Ramser (1959), who used a linear programming formulation in order to produce a near optimal solution with four routes to a problem of 12 drop points. Clarke and Wright (1964) generalized the Truck Dispatching Problem, which was introduced by Dantzig and Ramser (1959), to the Vehicle Routing Problem. It is now more than 50 years since the publication of these seminal works and the research in the field has an explosive growth. Relevant research has led to the development of new theories in VRP and other areas of combinatorial optimization. Ball et al. (1983), Christofides (1985), Golden and Assad (1986, 1988), and Ball (1995) presented distinguished research work for the VRP. Laporte and Osman (1995) have published a bibliography since 1995, containing over 500 articles on vehicle routing. The VRP, however, focuses on specific constraints imposed either by the customer or by the resources used to affect the delivery. When academic operations researchers address the complex Vehicle Routing Problem for scheduling the deliveries, they tend to fragment the various issues and tackle only a single issue in order to minimize the number of variables under consideration at any one time.

It is still encountered in our days, mainly in the domain of logistics and transport, and it is one of the most widely studied topics in the field of Operational Research. In the VRP, m vehicles, with identical capacities (Q), initially located at a central depot, are to deliver discrete quantities of goods to n customers, which are geographically diffused around the central depot. Concurrently, the aim of the VRP, beyond serving customers, is to minimize the travelled distance.

Due to the difficulty the VRP presents and because of its practical applications, many models have been created for solving the problem and many variants of the basic VRP have been compiled, with different parameters, leading to a different structure of the basic VRP. Firstly, the classical VRP is equivalent to the Capacitated

VRP (CVRP) in which, the capacity of the vehicle must not be exceeded (Lysgaard et al. 2004). However, there is another possibility that the vehicles do not have the same capacities which leads to the Heterogeneous Fleet VRP (HVRP).

Another important variant which was created a decade after the classical VRP was the Multi-Depot VRP (MDVRP) in which, the company has several depots from which it can serve its customers, while the objective is still to service all customers and minimize the number of vehicles and distance travelled (Montoya-Torres et al. 2015). At the same period, the Stochastic VRP (SVRP) was created, where, the customers, the demand of each customer or service and travel times are random.

In addition, in the VRP with Pick-up and Delivering (VRPPD), goods are transported, not only from the central depot to customers, but in the opposite direction too. Hence, in VRPPD it is necessary to take into account that the total delivery load and the total pickup load should both fit into the vehicle (Wassan and Nagy 2014). Furthermore, due to the need for specific arrival time information and for better customer services, an extra restriction to the VRP was added, time windows. The VRP with Time Windows (VRPTW) is the same with the classical VRP, but constrained by a time interval within which the customers have to be supplied (Beheshti et al. 2015).

Another variant worth mentioning is the Dynamic VRP (DVRP) in which customer requests, that are trips from an origin to a destination, appear dynamically (Pillac et al. 2013) or when data and information regarding to the VRP such as travel time, change dynamically. Another significant variant which takes into account time is the Time Dependent VRP (TDVRP) in which travel times change as time passes. The reason why this happens is due to traffic congestion. The factors which affect travel times are: (1) the location, and (2) the time of the day (Huang et al. 2017).

During the last decade, many researchers have tried, while solving the VRP, to minimize carbon (or fuel emissions) as carbon dioxide (CO2) emitted by trucks is the main greenhouse gas. In addition, the Green VRP has been strengthened due to the technical developments and the road traffic information which allows planning vehicle routes and schedules and taking time-varying speeds into account (Teng and Zhang 2016; Qian and Eglese 2016). In the same field belongs the Hybrid VRP where vehicles can work both electrically and with petroleum-based fuel (Mancini 2017).

VRP has numerous practical applications in the industry. Solving the Vehicle Routing Problem can assist companies in reducing their costs, as a big part of a product's price originates from transportation. Companies are trying to adopt VRP software solutions with appropriate algorithms in order to reduce transportation costs, mileages, and time spent while increasing customer services. Maybe the most challenging issue for companies is urban transportation, as traffic congestion often hinders the delivery of products on time.

The scope of the paper is to review and categorize research papers in the field of VRP for urban freight transportation and to provide a methodology for the analysis of the VRP research field and the categorization of the research papers. The results and methodology of the review can assist researchers in studying the research field,

as well as practitioners in understanding the various aspects of the VRP in order to support the design of software solutions.

In the next sections of this paper, we first record the papers which are related to the VRP for urban freight transportation through a specific research process in Sect. 2. Section 3 introduces the research methodology followed and presents the survey's main topics and some numerical results. In Sect. 4, we feature the trends of the VRP since the present paper will be used as an analysis for a research project of a Vehicle Routing and Scheduling Information System.

2 Literature Review Research Approach

The current paper followed a systematic literature review adopted from the work (Alexander et al. 2014; Eksioglu et al. 2009; Braekers et al. 2016). After the need for a review is identified, a research protocol is appointed, selection and de-selection criteria defined, and then papers assessed and outcomes composed. It aims at an efficient and well-recorded coverage which includes only necessary variants and may lead to potential developments and applications.

The Scopus database was used for an advanced search in accordance with Table 1. The aim of the present paper is to identify papers relevant to the VRP for urban freight transportation which are analyzed for featuring the last decade's trends. Every paper was studied thoroughly in order to achieve the right categorization which leads to an established statistical analysis.

The full list of papers extracted from the Scopus database, applying the search terms of Table 1 can be found in the following link (178 papers): https://goo.gl/1eRBfw.

Following the described process, including the application of deselection criteria of Table 1, 111 documents were selected and categorized according to the methodology proposed in the next section. This methodology produced a credible and essential data to an upcoming research project about VRP systems for urban freight transportation.

The final list of papers presented in this literature review after the application of deselection criteria of Table 1 can be found in the following link (111 papers): https://goo.gl/yoVrKQ.

Figure 1 shows the distribution of reviewed papers by journal. It is worth noting that the first journal is European Journal of Operational Research which accounts 13 papers. 17 journals account more than one paper (2–13 papers) and in total 65 papers which means 60% of the reviewed publications. All other 46 journals published 1 paper in the last decade according to the implemented research protocol of Table 1.

Figure 2 shows the distribution by publication year of the 111 reviewed papers as well as all the articles that fell under our search criteria in Scopus database (Table 1). Year 2017 on this diagram includes papers published during the first quarter of the year. According to this graphic, it is obvious that there is an increasing trend, demonstrating the increasing interest in "urban freight transportation" research area.

Table 1 Research Protocol

Research protocol	Title: Set of "Vehicle Routing Problem" or "VRP"
Research variable:	Description
Databases:	Scopus: This is an international database, covering a wide range of peer-reviewed academic publications, estimated at 20.000
Publication type:	Peer-reviewed papers only: These represent work at a final stage of completion
Language:	English-only: This provides wide coverage, and there was no translation capability
Data range:	The limits were set from 2006 till the first quarter of 2017, in order to include papers published at least in the last decade
Search fields:	Search terms were applied to titles, abstracts, and keywords only
Search terms:	(TITLE-ABS-KEY("Vehicle Routing Problem" or "VRP" or "Vehicle Routing") AND PUBYEAR >2006) AND ("urban freight transport*" OR "urban freight" OR "goods movement" OR "urban logistics" OR "city Logistics") AND (LIMIT-TO (SRCTYPE,"j") OR LIMIT-TO (SRCTYPE,"k") OR LIMIT-TO (SRCTYPE,"b")) AND (LIMIT-TO (DOCTYPE,"ar") OR LIMIT-TO (DOCTYPE,"ch") OR LIMIT-TO (DOCTYPE,"re") OR LIMIT-TO (DOCTYPE,"bk")) AND (LIMIT-TO (SUBJAREA,"COMP") OR LIMIT-TO (SUBJAREA,"ENGI") OR LIMIT-TO (SUBJAREA,"DECI") OR LIMIT-TO (SUBJAREA,"BUSI")) AND (LIMIT-TO (LANGUAGE,"English")) The primary search terms are the phrases "Vehicle Routing Problem," "VRP," and "Vehicle Routing." After the first phase, the terms "urban freight transport," "goods movement," "urban logistics," "urban freight," and "city logistics" are added, and the papers are limited to those whose subject area is Computer Science, Engineering, Decision Sciences, or Business Management, excluding Mathematics as the paper is focused in information systems and not to mathematical formulations. The former queries are applied to acclaimed document sources such as Article, Book Chapter, or Review and their Source type limited to Journal, Book Series, or Books
Deselection criteria: Relevance to the research problem:	The deselection criterion is relevant to the research scope: To explore applied solutions of the various VRP variants for urban freight transportation. Some papers are clearly irrelevant based on the title or the abstract while others are also deemed irrelevant after a more thorough review, as they refer to waste collection, cash in transit, truck and trailer, battery swap stations, location routing, food distribution, and parcels and documents. For the remaining papers, the full text was reviewed in order to achieve a detailed categorization

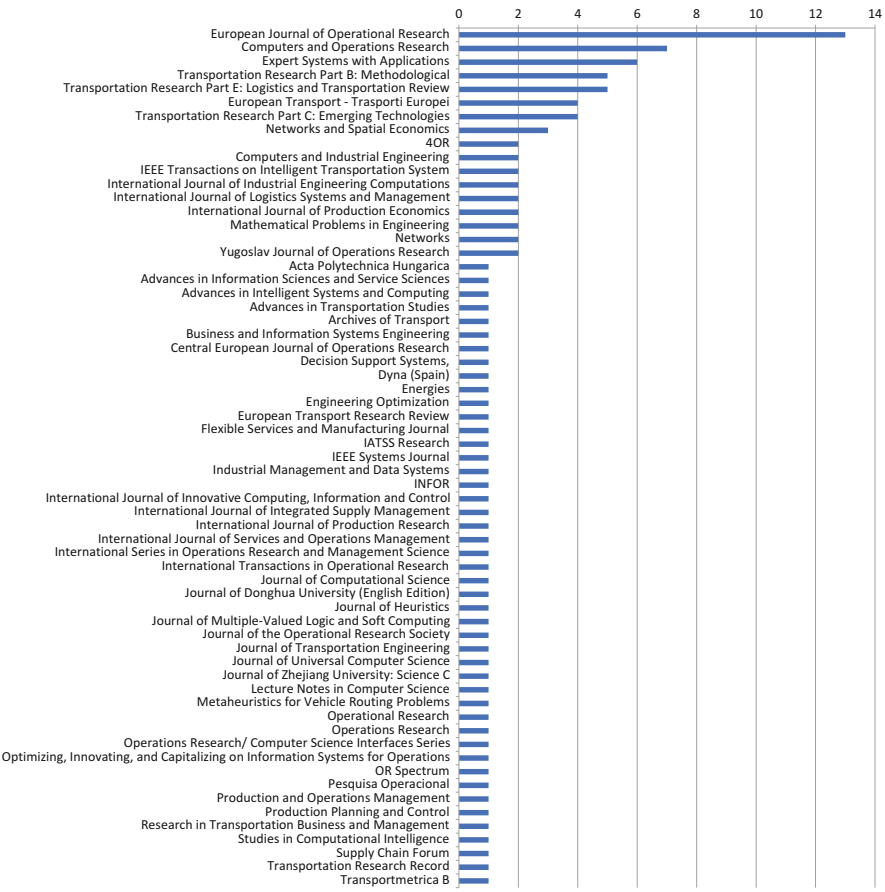


Fig. 1 Distribution of reviewed papers by journal

3 Methodology and Results

3.1 The Methodology for Analysis and Taxonomy of the Papers

The VRP is an extensively analyzed problem, which finds significant application in urban freight transportation. The delivery of products from distribution centers to several drop points within the cities faces difficulties such as traffic congestion, which lead to delays in the scheduled routes of vehicles. Hence, companies are seeking for software and systems which can reduce or ideally eliminate the obstacles, and schedule reliable routes, taking simultaneously more than one parameters of the classical VRP into account. Besides software solutions, mainly research papers propose and develop algorithms for solving specific variants of the VRP, without

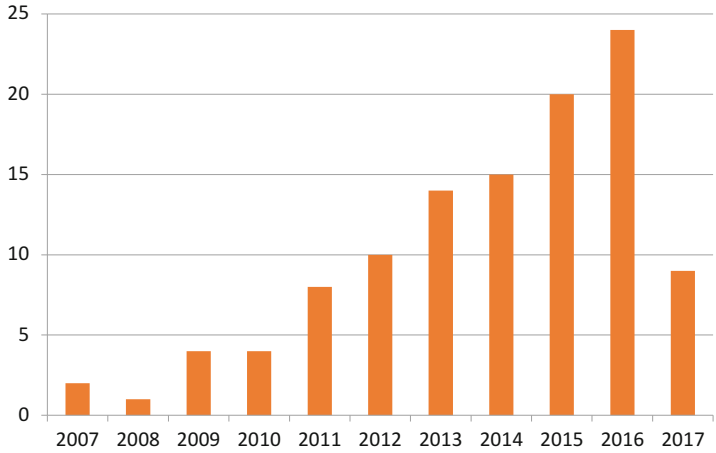


Fig. 2 Distribution of reviewed papers by year

offering complex solutions. The evaluation, primarily of software and secondarily of individual algorithms, is important to be implemented through case studies as they help in understanding real systems and taking reliable outcomes.

Consequently, the conducted survey is exhibited in three dimensions. The first dimension includes the different variants of the VRP, the second dimension consists of the algorithms used for the solution of these variants of the VRP, and the third dimension, the paper type.

Figure 3 depicts the methodology followed for the analysis of the finally selected 111 papers, including the three dimensions. These dimensions are appropriate for a systematic analysis of the research field of the VRP, in cases of applied projects for the design of specifications of VRP software, its development and implementation. Software solutions for VRP usually need adaptation and additional development in order to meet a specific company's needs. The proposed methodological approach and the research results can assist such projects in defining the VRP, selecting appropriate algorithms, and assessing the suitability of applied methods in similar cases. The three dimensions of the methodological approach are described in the following sections.

Dimension 1: VRP Variants

The most common variants of VRP include the Capacitated VRP (CVRP), the VRP with Time Windows (VRPTW), the Time Dependent VRP (TDVRP), the Green VRP, the Multi-Depot VRP (MDVRP), and the Multi-Echelon VRP (MEVRP). Besides these most common VRP variants, which are presented in the introduction, there are few more that have not yet been presented. The Two-Echelon VRP, originating from the Multi-Echelon VRP, involves two levels of routing problems,

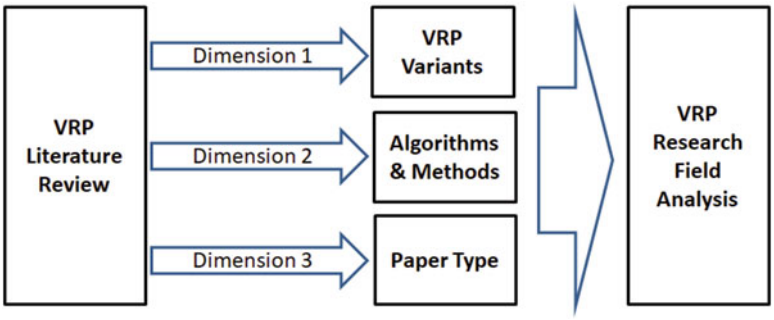


Fig. 3 Methodology for the analysis of the VRP research field

as satellites are supplied from the depot and the customer demands are transported from satellites (Baldacci et al. 2013). In the VRP with Satellite Facilities (VRPSF), satellites are used to replenish vehicle during a route, giving the opportunity to drivers making deliveries until the end of their shift, without necessarily returning to the central depot (Bard et al. 1998). A VRP which is similar to the VRPSF and gathers interest especially for city logistics is the Multi-Trip (MTVRP), where each vehicle can go back to the depot, be reloaded, and commence another trip, making several trips during the working day (Cattaruzza et al. 2014). As regards to the Multi-Depot VRP, every customer is visited exactly once by a vehicle based at one of several depots and each route must start and end at the same depot (Crevier et al. 2007; Renaud et al. 1996).

A very important variant which is close to the TDVRP is the Real-Time VRP where a re-optimization procedure runs in order for new routes to be identified, every time, travel time between a pair of nodes is updated (Okhrin and Richter 2008). Furthermore, Real-Time Traffic Data can be used to develop routing strategies that tend to minimize the time spent due to traffic congestion and consequently costs, while maximizing customer services (Kim and Lewis 2004).

In the Periodic VRP (PVRP) a set of customers has to be visited on a given time horizon, once or several times (Angelelli and Speranza 2002). In the Split Delivery VRP (SDVRP), the assumption that a customer must be visited by one vehicle only is relaxed and the objective is still to find the routes which minimize the cost (Archetti and Speranza 2012). Finally, the Open VRP differs from the classical VRP, as vehicles are not required to return to the depot after the end of the route (Sariklis and Powell 2000).

Dimension 2: Algorithms and Methods

In spite of the extensive research of the Vehicle Routing Problem, there is no single algorithm that optimally solves every problem, even no single algorithm for a specific VRP variant. Algorithms and methods have been developed for optimally

solving certain classes of vehicle routing and scheduling problems, tackling only a single issue in order to minimize the number of variables under consideration at any one time. Practical real-life applications of Vehicle Routing Problems offer heuristic or metaheuristic algorithms as solutions while there are some exact approaches in the literature.

Many researchers have faced most of the presented VRP variants in search of a solution. The methods they propose are Heuristic, Metaheuristic, Exact, and Simulation. Exact optimization methods may guarantee finding an optimal solution but due to the fact that the VRP is an NP Hard Problem only a few problems can be solved. On the contrary, heuristic and metaheuristic optimization methods may not guarantee that an optimal solution is found, but they provide an adequate solution that guides the search for the immediate goals, in less time (Rothlauf 2011). Simulation methods apply what-if analyses, for generating multiple scenarios, selecting the best one and using these in the main algorithms (Alexander et al. 2014).

Dimension 3: Paper Type

Besides the VRP variants and the kind of methods researchers present, it is important to estimate the number of papers (1) conducting a literature review or survey which means broader theoretical knowledge, (2) the papers proposing and developing algorithms or even these which describe or present another existing algorithm or any other solution, (3) the papers which have computational results and test data, and (4) the papers which present works on software and systems development which is a focal point in our research. This categorization of the papers is useful in order to distinguish the papers that are useful for assisting practical applications of the VRP algorithms, especially in freight transportation in cities.

3.2 Research Results

According to the categorization of 111 papers, the CVRP proves to be the most common variant of the VRP in urban freight transportation, followed by the VRPTW (Table 2). Many of the other VRP variants have lately been studied such as the Green and Hybrid VRP due to the need for minimization of carbon and fuel emissions or the TDVRP, the DVRP and the Real Time VRP, where new technologies offer better feasible solutions. In the research of Braekers et al. (2016), conducted for articles published till the middle of 2015, it is crucial to stress that the MEVRP and the VRPSF were not included, but have since attracted research interest. Other VRP variants such as the VRPTW, the MDVRP, the SVRP, and HVRP are producing similar results.

Table 2 lists all the VRP variants identified in the analyzed papers. As presented in this table, about 40% of the papers dealing with urban freight transportation examine Capacitated VRP and VRP with Time Windows. Many of the papers

Table 2 VRP variant

Variant	Number of articles (total = 111)	Relative percentage
Capacitated VRP	47	42.24%
VRP with Time Windows	42	37.84%
Green and Hybrid VRP	27	24.32%
Time Dependent VRP	21	18.92%
Multi-Echelon VRP and VRP with Satellite Facilities	15	13.51%
Multi-Depot VRP	11	9.91%
Stochastic VRP	10	9.01%
Dynamic VRP	9	8.11%
VRP with Heterogeneous Fleet	9	8.11%
Multi-trip VRP	8	7.21%
Real-Time VRP and VRP with Real-Time Data	8	7.21%
Pickup and Delivery VRP and VRP with Backhauls	7	6.31%
Periodic VRP	3	2.70%
Split Delivery VRP	2	1.80%
Open VRP	1	0.90%

examined deal with more than one VRP variants in order to solve real-life problems. That is the reason why in Table 2, the sum of the number of articles is more than 111. For example, 21 papers examine Time Depended VRP and Stochastic VRP, in parallel. Time Depended VRP is the problem which is combined with 15 other variants. Table 3 analytically presents how the variants of VRP are combined. Capacitated VRP (nr. 7 variant) is the most examined problem, and it is combined with all other variants (in 85 papers). VRP with Time Windows (nr. 4 variant) is combined with all the other variants in 63 papers totally. Dynamic VRP (nr. 1 variant) is combined with just 4 other variants in 7 papers totally.

In Table 4, the methods for solving VRP are presented, indicating that Heuristic and Metaheuristic algorithms are used more often to find solutions of the problem. Exact methods are developed less often due to their computational complexity while simulation methods are also less common in the literature. It is a logical outcome that heuristic and metaheuristic methods are used more often than the rest of the methods, mainly because the VRP is an NP Hard Problem. It should be noted the sum of Relative Percentages does not equal 100% due to the fact that many of the examined papers do not propose any solution but conduct a review or a survey.

In Table 5, the correlation between VRP variants and solution methods is presented. It is obvious that the two most common VRP variants, CVRP and VRPTW, are solved primarily with heuristic methods and secondarily with metaheuristic methods. On the other hand, the TDVRP and the Green and Hybrid VRP, which have been studied mostly over the last decade, are more often solved with metaheuristic approaches. This means that a trend seems to be developing for the application of

Table 3 Combinations of VRP variants (number of papers)

Variant	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	x			1			3				2				1
2		x		8	1	2	10	1	1	1	2	2	1		5
3			x	4	1		10					1	1	1	3
4				x	3	5	21	4	2	1	4	6	2	1	1
5					x	1	6		1		1		2		
6						x	3	1	1	1	1	1			
7							x	4	2	2	7	5	10		2
8								x	1	1		1			
9									x	1	1	1			
10										x		1			
11											x		1		
12												x			
13													x		
14														x	
15															x

Table appendix: VRP variants	
1	Dynamic VRP
2	Time Dependent VRP
3	Green and Hybrid VRP
4	VRP with Time Windows
5	Multi-trip VRP and VRP with Satellite Facilities
6	Multi-Depot VRP
7	Capacitated VRP
8	Pickup and Delivery VRP and VRP with Backhauls
9	Split Delivery VRP
10	Periodic VRP
11	Stochastic VRP
12	VRP with Heterogeneous Fleet
13	Multi Echelon VRP
14	Open VRP
15	Real-Time VRP and VRP with Real-Time Data

metaheuristic methods in solving the VRP. In regard to the simulation method, it is used mainly in the calculation of travel times, which is an important parameter for solving most of the VRP variants.

Most papers deal with more than one variants of the VRP each time. Hence, when the author of each paper solves a specific VRP with multiple variants, we consider that the method applied solves all these variants. That is the reason why in Table 5, the sum of each method differs from the data in Table 4.

Figure 4 depicts this relation between methods used in the literature and the VRP variant they deal with. Capacitated VRP, VRP with Time Windows, Time dependent

Table 4 Applied methods

Applied method	Number of articles (total = 111)	Relative percentage
Heuristic	41	36.94%
Metaheuristic	30	27.03%
Exact	5	4.50%
Simulation	5	4.50%

Table 5 Methods per VRP variant

Variant	Heuristic	Metaheuristic	Exact	Simulation
Capacitated VRP	20	12	3	2
VRP with Time Windows	20	11	1	3
Green and Hybrid VRP	5	8	0	1
Time Dependent VRP	5	6	0	0
Two Echelon VRP and VRP with Satellite Facilities	8	4	2	1
Multi-Depot VRP	4	4	0	1
Stochastic VRP	3	1	0	2
Dynamic VRP	3	0	0	0
VRP with Heterogeneous Fleet	3	4	0	0
Multi-trip VRP	4	2	1	1
VRP with Real-Time Traffic Data	2	0	0	0
Pickup and Delivery VRP and VRP with Backhauls	3	3	0	0
Periodic VRP	2	1	1	0
Split Delivery VRP	2	1	o	1
Open VRP	0	1	0	0

VRP and VRP with Real-Time Traffic Data are the most relevant variants for urban freight transportation problems and in Fig. 4 is obvious that the most commonly used methods for such problems are heuristics and metaheuristics.

Table 6 indicates that most papers present Computational Results or Test Data due to the fact that a significant percentage of papers are relevant to Algorithms (whether they Propose/Develop or Present/Describe Algorithms), Case Studies, and Real-World Industrial Examples. Many Algorithms, and especially those who are proposed and developed, find application and tested in case studies, in order to be evaluated and draw conclusions from real problems. Furthermore, in many papers, time and traffic data of cities are received and are used in companies, in order to test and decide if the software and algorithm solutions they use are trustworthy. Literature Reviews or Surveys follow, as many researchers approach the VRP through a theoretical viewpoint. Finally, Software or Systems are rare, as few researches lead to such applications.

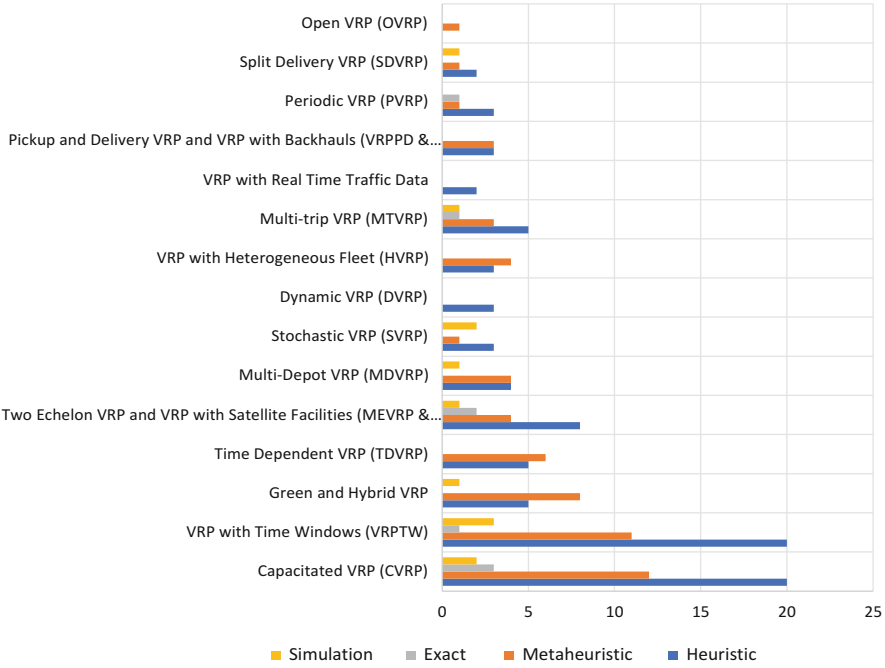


Fig. 4 Methods used per VRP variant

Table 6 Paper types

Paper type	Number of articles (total = 111)	Relative percentage
Computational Results or Test Data	80	72.07%
Propose/Develop Algorithm or Present/Describe Algorithm	59	53.15%
Case Study and Real-World Industrial Examples	27	24.32%
Literature Review or Survey	22	19.82%
Software or System	5	4.50%

4 Conclusions

A considerable number of papers have been written about the VRP for urban freight transportation, analyzing many different variants of it. Accordingly, many methods have been proposed for solving all the VRP variants. However, few research projects for the VRP have led to the real-life application of Decision Support Systems (DSS) or other Management Systems (MS). The papers of Zeimpekis et al. (2007, 2017) and Sun and Hu (2012) refer to Vehicle MS and DSS that endorse logistics managers by monitoring the status of deliveries in real time and suggesting optimal re-routing strategies whenever needed due to uncontrollable external events that

cause perturbations in the schedule. The paper of Smirlis et al. (2012) presents a model which evaluates software products through the total performance/price criterion and proposes the proper solution as the selection of these systems has become increasingly difficult for decision makers due to a large number of software products available and the great variety of features and capabilities they offer. There are systems which are based on GIS (Geographical Information Systems) such as those described by Sicilia-Montalvo et al. (2013). Authors are also reporting the need for ITS (Intelligent Transportation Systems), offering the confrontation of high costs and environmental impact which is important especially in urban areas. Due to the need for lower costs, lower carbon emissions, and higher customer services, DSS and MS can lead to sufficient and feasible solutions.

This paper presents the results of an extensive literature review of the VRP for urban freight transportation. An initial group of 178 papers extracted from Scopus database, and it was further shorted to a group of 111 relevant papers, applying deselection criteria. Papers which are irrelevant with applied solutions of the various VRP variants for urban freight transportation were excluded from the analysis. A link to all 111 examined papers is given and the analysis is presented in six tables and four figures. This review offers a summary of the latest knowledge of this research field, and it can enhance the adaptation of appropriate methods in software solutions for freight transportation in cities.

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